

## Activity-2

1) CCTV image contains salt-and-Pepper noise

Technique: Median Filter

It is used to remove salt-and-Pepper noise from CCTV images. It effectively removes impulse noise while preserving image edges.

2) Satellite image corrupted by Gaussian noise

Technique: Gaussian filter

It reduces Gaussian noise by smoothing the image. It is commonly used for satellite images affected by sensor noise.

3) Medical x-ray has random noise but fine edges must be preserved.

Technique: Bilateral filter

It removes random noise while preserving fine edges. It is ideal for medical x-ray images where edge details are important.

4) Scanned document has black & white dots scattered randomly

Technique: Median filter

It removes black & white dots from scanned documents. It clears the image without affecting text quality.

5) Low-light image shows heavy grain

Technique: Non-local Means (NLM) filter

It reduces heavy grain in low-light

images are clearer and less noisy images.

6) Foggy road image has very low contrast

Technique: Histogram Equalization

It improves the contrast of foggy road images. It makes hidden details more visible by spreading intensity values.

7) Underwater image appears dull and washed out.

Technique: Contrast stretching.

It enhances dull underwater images by expanding the intensity range. This makes the image brighter and clearer.

8) Chest X-Ray shows Poor Visibility of internal structure

Technique: Adaptive Histogram Equalization [CLAHE]

It improves the visibility of internal structures in chest X-rays. It enhances local contrast without amplifying noise excessively.

9) Night-time traffic image lacks intensity variation

Technique: Histogram Equalization

It increases intensity variation in night time traffic images. It improves overall brightness and contrast.

10) Grayscale image uses only a narrow intensity range

Technique: Contrast Stretching

It expands the narrow grayscale intensity range. This improves image contrast and makes details easier to see.